

Sharing Resources Across Diverse OSs with Samba

Welcome to the world of Samba. Being software that can be run on multiple platforms such as Windows, UNIX, Linux, etc, it enables you to share your files and printers between systems running various OSs. It works as an alternative to NFS.



Samba is an open source network file system. It allows access to shared resources across the different OSs such as UNIX, Linux, IBM System 390, OpenVMS, etc. It is an open source implementation of Microsoft's SMB (Server Message Block)/CIFS (Common Internet File System) protocol.

The main components of Samba

Samba is made of three daemons (smbd, nmbd and winbindd). There are two services (smb and winbind) that control the behaviour of daemons.

- **smbd:** The server message block daemon provides sharing and printing services over the network. It also controls resource locking, user authentication and data sharing through the SMB protocol. The Samba server listens for SMB traffic on the default TCP ports, 139 and 445. The smb service controls the behaviour of the smbd daemon.
- **nmbd:** This provides NetBIOS names over the network. It allows the browsing protocols to make up the Windows Network Neighborhood view. The Samba server listens for NMB traffic on the default UDP port 137. The smb service also controls the behaviour of the nmbd daemon.
- **winbindd:** This resolves user and group information on a server running Windows NT 2000 or Windows Server 2003. It renders Windows user/group information in a format that is understandable by UNIX platforms.

A Microsoft RPC call, the name service switch (NSS), and pluggable authentication modules (PAM) allow this daemon to achieve this. This allows Windows NT domain users to access a UNIX machine as UNIX users. The winbind service controls the behaviour of the winbindd daemon. It doesn't require the smb service to be started. It is a client side service, which is used to connect to Windows users.

Samba also contains several important files.

- **smb.conf:** This basic configuration file can be edited directly with a text editor or via a utility program like SWAT or Webmin.
- **smbpasswd:** This contains the names of users to which smbd will refer when it is time to enforce user-level access control.
- **smbusers:** This provides the names of Samba's administrators and of users with specialised Samba access.

Samba's features

The primary features of Samba are shown in Figure 2.

The other features are:

- Establishes shares on UNIX that are accessible to Windows users
- Shares printers
- Creates a naming scheme so one can use user-friendly names rather than IP addresses
- Makes UNIX a Windows Internet Naming Service (WINS) server